Practical No: 5

Aim: Selenium Server (Grid) Configuration and Remote Execution Tool: Selenium Server (Grid) Install and configure Selenium Server in Hub-Node architecture. Execute a test script on multiple browsers using Remote WebDriver and compare cross-browser results.

Step I:

Requirements

Java JDK 11 or above

Selenium Server (latest JAR)

Chrome browser

Firefox browser

ChromeDriver and GeckoDriver (optional for older versions)

Eclipse/IntelliJ IDEA

Selenium Java library

DOWNLOAD JAVA JDK

Java JDK 11 or above

Step 1: Download Selenium Server

Download the Selenium Server JAR from the official Selenium website.

AFTER DOWNLOAD GO TO C DRIVE CREATE SELENIUM FOLDER IN C DRIVE

Step 3: Configure Selenium Hub

OPEN COMMAND PROMPT

cd C:\selenium

C:\selenium>**java -jar selenium-server-4.46.0.jar info**

C:\selenium>**java -jar selenium-server-4.46.0.jar hub**

```
C:\Users\user>cd C:\selenium

C:\Selenium>java -jar selenium-server-4.46.0.jar info

Info
====

You can find out more about topics concerning Selenium Grid by running the
"info" command followed by:

* config - Help on how to configure Selenium
* tracing - Help on how distributed tracing works with selenium
* security - Help on how how to secure communications to and within selenium
* sessionmap - Help on different types of sessionmap and how to start them.
* distributor - Help on Distributor backends (local, Redis) and how to start
them.
* kubernetes - Help on running browser sessions as Kubernetes Jobs.

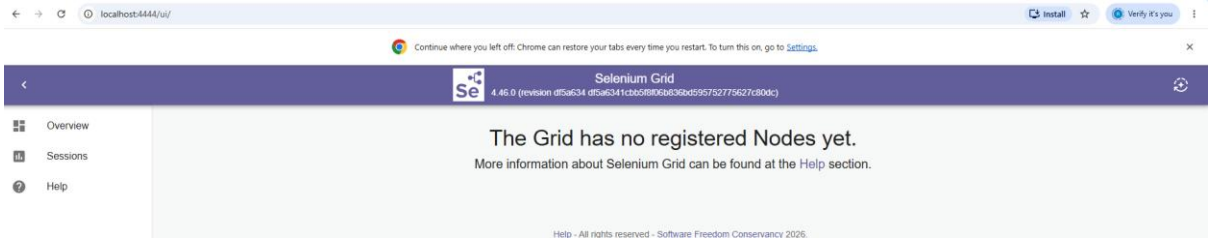

Each topic will give you enough information to help you get started, and contains
some pointers on our site to provide more complete information.



C:\Selenium>java -jar selenium-server-4.46.0.jar hub
11:54:25.095 INFO [LoggingOptions.configureLogEncoding] - Using the system default encoding
11:54:25.111 INFO [OpenTelemetryTracer.createTracer] - Using OpenTelemetry for tracing
11:54:25.426 INFO [BoundZmqEventBus.<init>] - XPUB binding to [binding to tcp://*:4442, adv
43]
11:54:25.443 INFO [ZmqUtils.configureHeartbeat] - ZMQ XPUB socket heartbeat configured: int
11:54:25.443 INFO [ZmqUtils.configureHeartbeat] - ZMQ XSUB socket heartbeat configured: int
11:54:25.458 INFO [UnboundZmqEventBus.<init>] - Connecting to tcp://192.168.0.111:4442 and
11:54:25.477 INFO [ZmqUtils.configureHeartbeat] - ZMQ SUB socket heartbeat configured: inte
11:54:25.477 INFO [ZmqUtils.configureHeartbeat] - ZMQ PUB socket heartbeat configured: inte
11:54:25.486 INFO [UnboundZmqEventBus.<init>] - Sockets created
11:54:26.490 INFO [UnboundZmqEventBus.<init>] - Event bus ready
11:54:27.374 INFO [Hub.execute] - Started Selenium Hub 4.46.0 (revision df5a634 df5a6341cbb
```

Search on chrome Started Selenium Hub

http://localhost:4444



Step 4: Configure Selenium Node

A Node is a machine/browser where tests are executed.

Start Chrome Node

Open another Command Prompt window.

Go to Selenium folder:

cd C:\Selenium



Start Node:

java -jar selenium-server-4.46.0.jar node

```
C:\Selenium>java -jar selenium-server-4.46.0.jar node
11:59:54.813 INFO [LoggingOptions.configureLogEncoding] - Using the system default
11:59:54.828 INFO [OpenTelemetryTracer.createTracer] - Using OpenTelemetry for tra
11:59:54.954 INFO [UnboundZmqEventBus.<init>] - Connecting to tcp://0.0.0.0:4442 a
11:59:54.980 INFO [ZmqUtils.configureHeartbeat] - ZMQ SUB socket heartbeat configu
11:59:54.985 INFO [ZmqUtils.configureHeartbeat] - ZMQ PUB socket heartbeat configu
11:59:54.985 INFO [UnboundZmqEventBus.<init>] - Sockets created
11:59:55.993 INFO [UnboundZmqEventBus.<init>] - Event bus ready
11:59:56.171 INFO [NodeServer.createHandlers] - Reporting self as: http://192.168.
11:59:56.440 INFO [NodeOptions.getSessionFactories] - Detected 16 available proces
11:59:56.440 INFO [NodeOptions.discoverDrivers] - Looking for existing drivers on
11:59:56.440 INFO [NodeOptions.discoverDrivers] - Add '--selenium-manager true' to
11:59:56.759 WARN [SeleniumManager.lambda$runCommand$0] - Unable to discover prope
11:59:56.834 WARN [SeleniumManager.lambda$runCommand$0] - Unable to discover prope
11:59:56.853 INFO [NodeOptions.report] - Adding Chrome for {"browserName": "chrome
11:59:56.853 INFO [NodeOptions.report] - Adding Edge for {"browserName": "Microsof
11:59:56.853 INFO [NodeOptions.report] - Adding Firefox for {"browserName": "firef
11:59:56.853 INFO [NodeOptions.report] - Adding Internet Explorer for {"browserNam
11:59:56.884 INFO [Node.<init>] - Binding additional locator mechanisms: relative
11:59:57.294 INFO [NodeServer$2.start] - Starting registration process for Node ht
11:59:57.294 INFO [NodeServer.execute] - Started Selenium node 4.46.0 (revision df
11:59:57.310 INFO [NodeServer$2.lambda$start$3] - Sending registration event...
11:59:58.054 INFO [NodeServer.lambda$createHandlers$1] - Node has been added
```

Step 5: Add Firefox Node (Optional)

If Firefox is installed:

Open another Command Prompt.

Run:**java -jar selenium-server-4.46.0.jar node --browser firefox**

```
C:\Users\user>cd C:\selenium

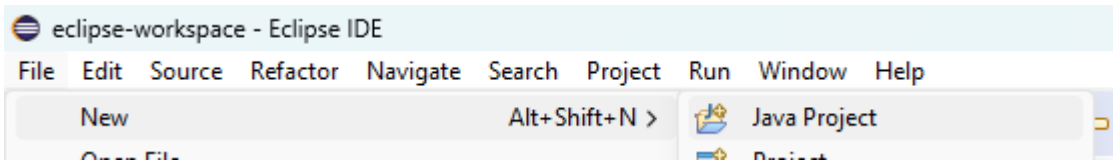
C:\Selenium>java -jar selenium-server-4.46.0.jar node --browser firefox
Was passed main parameter '--browser' but no main parameter was defined in your arg class
Usage: node [options]
  Options:
    --allow-cors
      Whether the Selenium server should allow web browser connections from
      any host
      Default: false
    --bind-bus
```



Step 7: Create Selenium Project

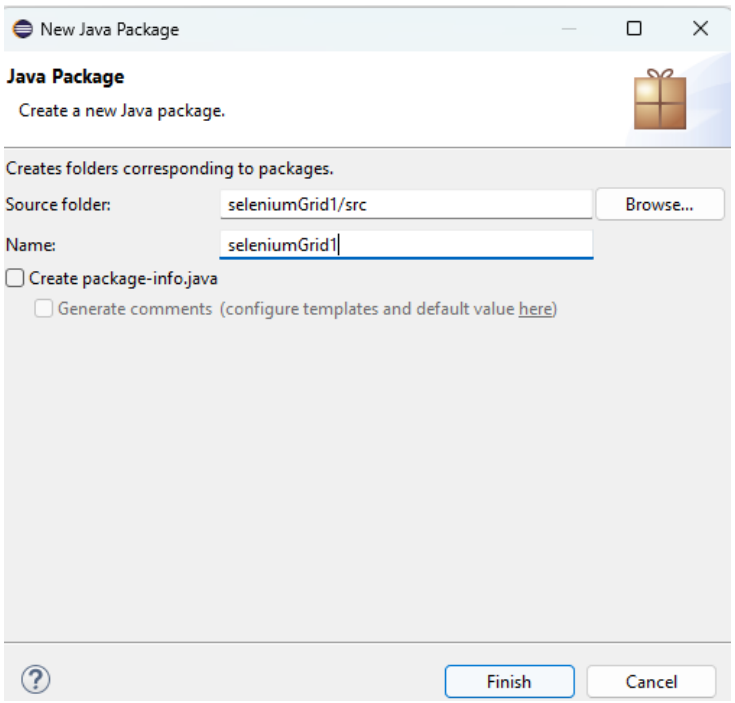
OPEN ECLIPS:FILE NEW JAVA PROJECT ADD PCKAGE CLASS FILE





Make package



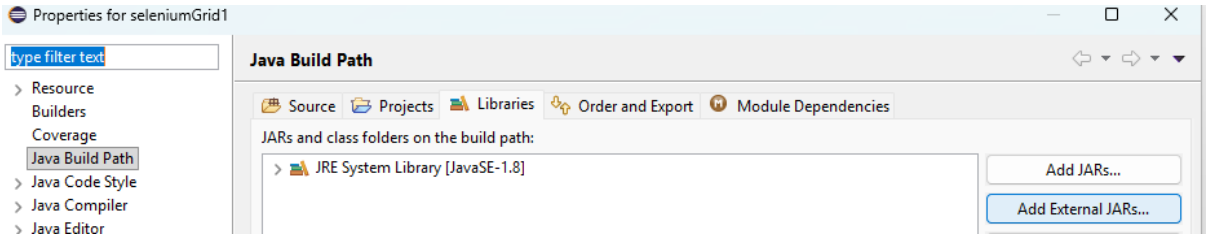


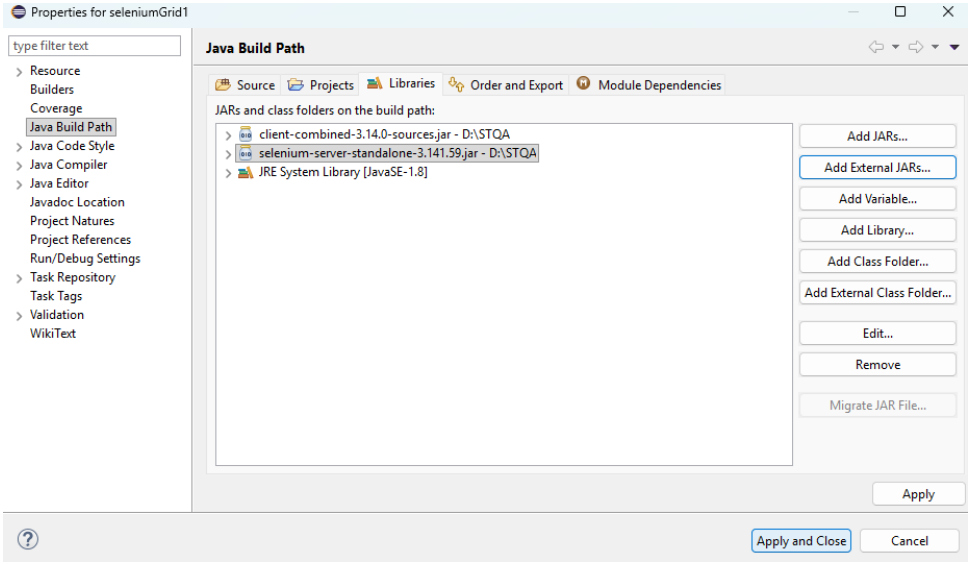
- **Make two classes**





- Add External Jars





Write this code in grid 1:

```
package seleniumGrid1;

import java.net.URL;

import org.openqa.selenium.WebDriver;

import org.openqa.selenium.chrome.ChromeOptions;

import org.openqa.selenium.remote.RemoteWebDriver;


public class grid1 {

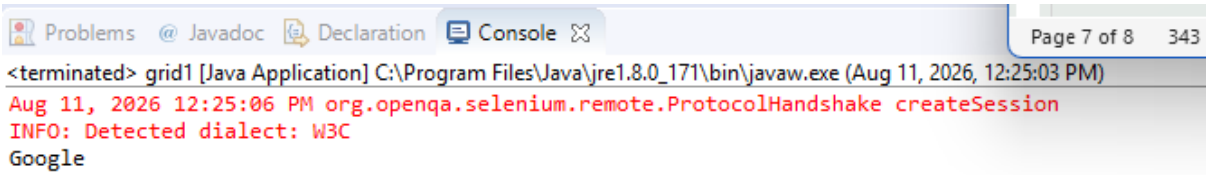
	public static void main(String[] args) throws Exception {

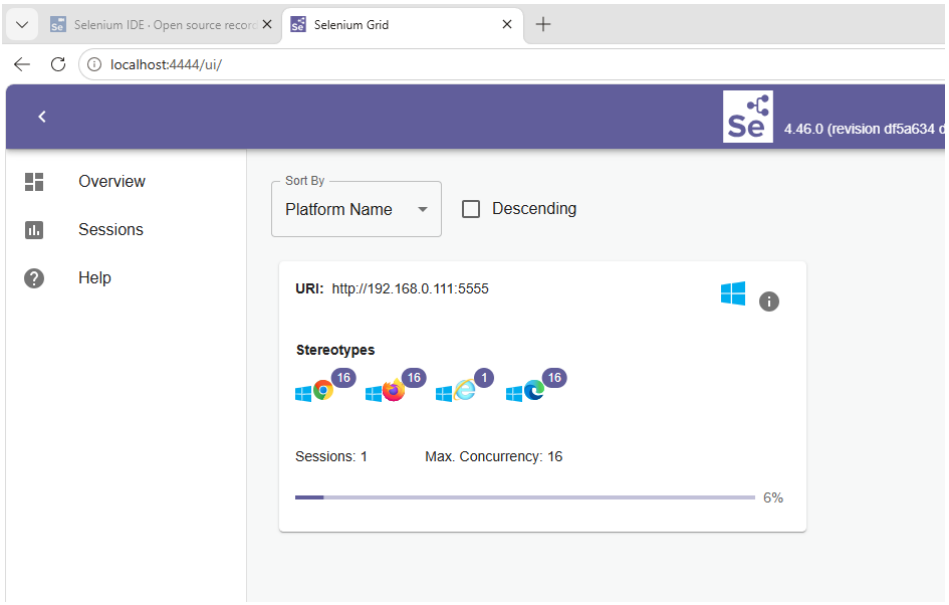
		ChromeOptions options = new ChromeOptions();

		WebDriver driver=new RemoteWebDriver(
```

```
                                    new URL("http://localhost:4444"),
                                    options);
            driver.get("https://www.google.com");
            System.out.println(driver.getTitle());
            driver.quit();
        }
}
```

The you run it will open google





code for grid 2:

```
package seleniumGrid1;

import java.net.URL;

import org.openqa.selenium.WebDriver;

import org.openqa.selenium.firefox.FirefoxOptions;

import org.openqa.selenium.remote.RemoteWebDriver;


public class grid2 {
```

```
    public static void main(String[] args) {
            try {
            // Connect to Selenium Grid
            URL gridURL = new URL("http://localhost:4444");
            // Set Firefox browser options
            FirefoxOptions options = new FirefoxOptions();
            // Create RemoteWebDriver
            WebDriver driver = new RemoteWebDriver(gridURL, options);
            // Open website
            driver.get("https://www.google.com");
            // Print page title
            System.out.println("Page Title: " + driver.getTitle());
            // Maximize browser
            driver.manage().window().maximize();
            // Wait for 5 seconds
            Thread.sleep(5000);
            // Close browser
            driver.quit();
        } catch (Exception e) {
            e.printStackTrace();
             }
    }
}
```

